

Systems Engineering in Public Programs - Course Map

Week	Date	Course focus	Expected Learning Outcomes	In-class Activity	At-home activity	
Week 1	Sep 04, 2025	Presentation Course	Get to know each other. Set expectations between you (the students) and me (the professor). Negotiation and communication skills. Self-assessment skills	Presentation of the class structure and expected learning outcomes.	Reading 2 resources + 1 video	Pre-assessment quiz *complete for full grade; Initial Feedback survey;
Week 2	Sep 09, 2025	Overview of the Models Modeling Equitable Systems	Identify the characteristics of Modelling Equitable System, Present an overview of Equitable OR framework Present characteristics of Equitable Systems	Mini-lecture on OR framework in social context and stakeholders' importance.	Assess a Model by its Equitable considerations	Discussion Posts (Online)
Week 2	Sep 11, 2025	Modeling Equitable Systems Case study	Learn the skill to extract information from multiple sources Understand and critically evaluate different sources content. Hone self-assessment skills	Mini-lecture on Models. Analysis of case study in 3 parts, class; in group and as a class discussion	Mini-lecture on how to read a Case study Project Case study material (part I)	Discussion Posts (Online)
Week 3	Sep 16, 2025	Multiple Objective Decision Making Main concepts: Multiple perspectives in decision-making Stakeholders	Stakeholders and their relation to decision-making Consider multiple perspectives in decision-making Design adequate questions to the problem based on stakeholders and expected outcomes.	Exercise in class – Stakeholders Self-explanation of exercise at home (posted) - Pass a problem activity with stakeholders and consequence of decision - Discussion of potential outcomes and impacts on each stakeholder (It will be part of the report, but discussion will be made in group)	Mini-lecture on different (non-obvious) stakeholders in decision Project Case study material (part II)	

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Week 3	Sep 18, 2025	Multiple Objective Decision Making Case study Main concepts: Consequences of decisions Consequences of different perspectives to modeling decisions in social settings.	Identify at least one unintended consequence of the modeling choices. Identify the trade-offs between different decisions and strategies. Offer at least one alternative modeling strategy. Adapt decision-model strategies to different contexts.	Discussion on Identity (Character building) 4S Activity – Decision outcomes in the real world (brainstorm, voting, discussion, voting)	Project Case study material (part II)	Post: 200 words post about what is the most important thing when listing assumptions for our models
Week 4	Sep 23, 2025	Data Envelopment Analysis Main concepts: Assumptions Data limitations Unintended Consequences			Thinking - Aloud Pair Problem-Solving Discussion of expected results Define final assumptions	
Week 4	Sep 25, 2025	Data Envelopment Analysis - Case study				Discussion Posts (Online)
Week 5	Sep 30, 2025	Presentations - Project Part I				Peer evaluation (online)
Week 5	Oct 02, 2025	Presentations - Project Part I				Peer evaluation (online)
Week 6	Oct 07, 2025	Presentation for multiple stakeholders - Project Part I				Peer evaluation (online)
Week 6	Oct 09, 2025	Public Health Preparedness and Response Model				
Week 7	Oct 14, 2025	Public Health Preparedness and Response Model - Simulation Activities/ Stakeholders				
	Oct 16, 2025	Interdiction and Attacker-Defender				At-home Exercises/ Jupyter Notebooks
Week 8	Oct 21, 2025	Interdiction and Attacker-Defender - Game: Spot the limitations				

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Week 8	Oct 23, 2025	Tutorial sessions - in person				Peer (in group) and Instructor
Week 9	Oct 28, 2025	Tutorial sessions - in person				Peer (in group) and Instructor
Week 9	Oct 30, 2025	Presentations - Project Part II				Peer evaluation (online)
Week 10	Nov 04, 2025	Presentations - Project Part II				Peer evaluation (online), report due
Week 10	Nov 06, 2025	Multiple Objective Decision Making				
Week 11	Nov 11, 2025	USA: Veterans Day, no classes				
Week 11	Nov 13, 2025	Data Envelopment Analysis				
Week 12	Nov 18, 2025	Public Health Preparedness and Response Model				Simulation Activities/ Feasibility
Week 12	Nov 20, 2025	Interdiction and Attacker-Defender				
Week 13	Nov 25, 2025	Multicriteria and Ranking Models				At-home Exercises/ Jupyter Notebooks
Week 13	Nov 27, 2025	Tutorial sessions - in person				Peer (in group) and Instructor
Week 14	Dec 02, 2025	Tutorial sessions - in person				Peer (in group) and Instructor
Week 14	Dec 04, 2025	Tutorial sessions - in person				Peer (in group) and Instructor
Week 15	Dec 06, 2025	Presentations - Project Part III				Peer evaluation (online)
Week 16	Dec 08, 2025	Presentations - Project Part III				Peer evaluation (online), report due